

Data Management Technologies

Unit 2 - Consolidation Task 5

Scenario Analysis and Use Cases

The chosen scenario is a sixth-form college with courses in 15 subjects. The college has over 500 students and 25 teachers. Each student is registered on 3 or 4 courses, and each teacher teaches 1 to 3 courses.

The database will also hold searchable data on parents and extra-curricular activities. This system enables staff to search for students by name, year group, and ID. It also provides a way for desired data to be compiled and printed while excluding sensitive personal data.

Lists of Entities, Attributes and Relationships

Entities

- Parents
- Students
- Courses
- Teachers
- Extra-curriculars

Attributes

STUDENTS	PARENTS	TEACHERS	COURSES	EXTRA-CURRICULARS
Student ID	Parent ID	Staff ID	Course ID	Activity ID
First name	First name	First name	Course title	Activity name
Last name	Last name	Last name	Course level	Activity notes
Linked parent (multivalued)	Linked student (multivalued)	Course ID (multivalued)	Course sessions (multivalued)	Activity schedule
Email address	Email address	Email address	Course description	Responsible staff
Address (composite field: line 1, line 2, and postcode)	Address (composite field: line 1, line 2, and postcode)	Activity ID (multivalued)	Target qualification	
Additional information	Phone number	Start date		
Preferred name		Department		
Date of birth		Phone number		
Year group		Employment type		

Notes on multivalued attributes:

A multivalued attribute is an attribute that can hold multiple values for a single entity.

Multivalued attributes in the scenario are:

- **Linked parents:** A student can have more than one parent registered (e.g. a mother and a father).
- **Linked students:** A parent can have more than one child registered (siblings).
- **Teacher's courses:** Some teachers could teach more than one course and extra-curricular activity.
- **Course sessions:** A course can have multiple scheduled sessions in the week.

Relationships

- Parents are linked to students
- Students are registered on courses
- Students are registered on extra-curriculars
- Courses are taught by teachers
- Extra-curriculars are moderated by teachers.

